# Hough-SIFT: Hough Pose Clustering for Learned Local Features with Semantic-Gated Voting

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## Abstract

Geometric verification of local feature matches is almost always done with RANSAC, whose success probability collapses as the outlier ratio grows. Lowe's Hough pose clustering is a deterministic alternative that is robust to outliers, but it requires every keypoint to carry a scale and an orientation, which learned detectors such as DISK and SuperPoint do not provide. We make Hough verification usable with orientation-free features through *pairwise similarity voting*: two correspondences determine a full similarity transform, so neighbouring pairs of matches cast the votes. Naive pairwise voting collapses under heavy contamination; we diagnose why (wrong matches produce absurd-scale votes that pile up in oversized location bins) and fix it with a scale prior and a mutual-spatial-neighbour rule. On HPatches (580 pairs) our verifier is on par with MAGSAC++ up to 90% injected outliers and clearly ahead at 95% (75.2% vs. 36.9% of pairs within 3 px with SIFT, 51.4% vs. 33.8% with DISK). We then study *semantic-gated Hough voting*, in which Mask R-CNN instance masks decide which matches may vote together and the instance identity becomes an extra Hough dimension. On a synthetic multi-object benchmark built from COCO, with design choices made on a held-out validation split, gating plus duplicate merging keeps object recall almost constant from 0% to 80% outliers (74.3% to 72.7% with DISK) while sequential RANSAC drops from 85.4% to 19.0%, and at 80% outliers it produces 4$\times$ fewer wrong hypotheses than semantic RANSAC at 17$\times$ lower cost. On clean scenes, plain RANSAC remains the best choice; we report this and the variants that did not work.

## 1 Introduction

Matching local features between two images is a basic step in object recognition, image stitching, visual localisation and structure from motion. Descriptor matching always returns a mixture of correct and incorrect correspondences, so a *geometric verification* step must decide which matches are consistent with a single physical motion. The standard tool is RANSAC Fischler and Bolles [1981] and its modern variants such as MAGSAC++ Barath et al. [2020]: repeatedly sample a minimal set of matches, fit a model and count agreeing matches. RANSAC's weakness is well known. A minimal sample of four matches is outlier-free with probability $(1-\varepsilon)^4$ for outlier ratio $\varepsilon$, which is $10^{-4}$ at $\varepsilon = 0.9$, so the number of required iterations explodes exactly in the hard cases.

Lowe's object recognition system Lowe [2004] used a different strategy: every SIFT match votes for an object pose (scale, rotation and position) in a coarse Hough accumulator Duda and Hart [1972], Ballard [1981]. Correct matches agree and form a peak; incorrect matches spread their votes thinly across pose space. Voting is deterministic, uses each match once and degrades gracefully with outliers. Hough verification has, however, largely disappeared from modern pipelines built on learned features. Benchmarks of learned matchers still verify with RANSAC Yu et al. [2024]. One reason

Course project report, Computer Vision, IIT Jodhpur, November 2026.

is structural: a single SIFT match can vote because each SIFT keypoint carries its own scale and orientation. Learned detectors such as DISK Tyszkiewicz et al. [2020] and SuperPoint DeTone et al. [2018] output only a 2-D location, so a single match no longer determines a pose.

A second weakness of classical Hough voting appears in cluttered scenes: *any* two matches may vote together, so a pair spanning two different objects casts a meaningless vote, and two objects with similar motion can merge into one cluster. Semantic segmentation knows which pixels belong to which object, which is exactly the information needed to prevent this.

**Contributions.**

1. **Pairwise similarity voting for orientation-free features.** We adapt Lowe-style Hough verification to learned detectors by letting pairs of neighbouring matches vote, since two correspondences determine scale and rotation. We identify why naive pairwise voting collapses under heavy contamination and fix it with a scale prior and a mutual-spatial-neighbour rule.
2. **Semantic-gated Hough voting.** Instance masks from an off-the-shelf segmenter decide which matches may vote together, and the instance identity becomes an additional Hough dimension, so votes from different objects never share a bin. A *soft* rule treats background as "unknown", and duplicate merging removes the redundant hypotheses that gating creates. We show that gating and voting are complementary: under heavy clutter the combination outperforms either component alone.
3. **Evaluation, including negative results.** An outlier-injection study on HPatches Balntas et al. [2017] against RANSAC and MAGSAC++ Barath et al. [2020], a sensitivity analysis of the neighbourhood size, and a $2 \times 2 \times 2$ ablation on a synthetic multi-object benchmark built from COCO Lin et al. [2014] with ground-truth motion for every object, with a separate validation split for design decisions. We also report the variants that failed and why.

## 2 Literature Review

**Hough-based geometric verification.** The Hough transform Duda and Hart [1972] and its generalisation to arbitrary shapes Ballard [1981] turn model fitting into peak finding in a parameter space. Lowe Lowe [2004] applied it to SIFT matches with a 4-D similarity pose space and coarse bins (30° orientation, factor 2 in scale, a quarter of the model size in location), followed by least-squares affine refinement. Hough pyramid matching Tolias and Avrithis [2011] relaxes the quantisation with a hierarchy of bin sizes, and the vote-and-verify strategy of Schönberger et al. Schönberger et al. [2016] combines voting with hypothesis verification for fast spatial verification in retrieval. All of these rely on per-keypoint scale and orientation.

**Robust estimation by sampling.** RANSAC Fischler and Bolles [1981] remains the default verifier. MAGSAC++ Barath et al. [2020] removes the need for a hand-tuned inlier threshold by marginalising over noise scales and is among the strongest classical estimators. Multi-model problems (several independently moving objects) are commonly handled by sequential RANSAC or energy-based methods such as PEARL Isack and Boykov [2012].

**Pairwise geometric consistency.** Pairs of correspondences carry more geometry than single ones. Spectral matching Leordeanu and Hebert [2005] scores pairwise consistency between matches, Li et al. Li et al. [2015] use pairwise geometric relations for large-scale object retrieval, and GMS Bian et al. [2017] rejects matches whose spatial neighbourhoods are not preserved between images. Our pairwise voting and our mutual-neighbour rule build on the same intuition, applied inside a Hough accumulator for learned features.

**Learned local features.** SuperPoint DeTone et al. [2018], D2-Net Dusmanu et al. [2019] and DISK Tyszkiewicz et al. [2020] learn detection and description jointly and outperform SIFT on many benchmarks; learned matchers such as LightGlue Lindenberger et al. [2023] further improve the matching step. These detectors provide keypoint locations without scale or orientation, and their downstream verification is almost always RANSAC Yu et al. [2024].

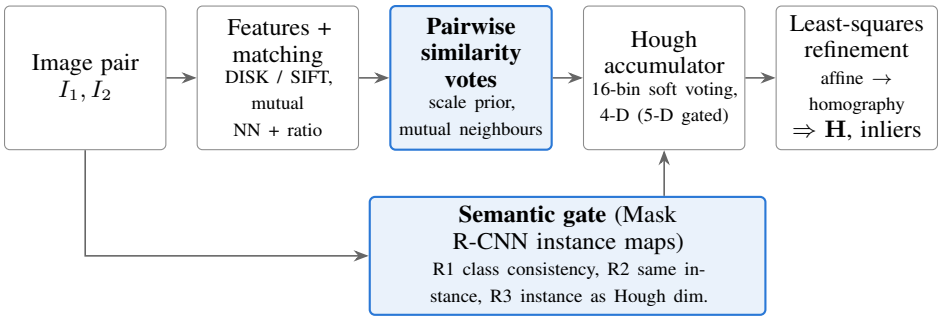



Figure 1: Overview of the proposed verification pipeline. Blue boxes are our contributions; the semantic gate is optional and only used for multi-object scenes.

**Segmentation-guided matching.** The closest prior work is the co-segmentation guided Hough transform of Chen et al. Chen et al. [2015], which uses unsupervised co-segmentation of an image pair to select relevant voters in a Hough space of homographies. SGAM Zhang and Zhao [2023] first matches semantic areas and then points inside matched areas; MESA Zhang and Zhao [2024] replaces semantic labels with Segment Anything Kirillov et al. [2023] masks and reports that SGAM degrades with inexact semantic labels. Our work differs from these in three ways: we use *instance* segmentation from a pretrained detector (Mask R-CNN He et al. [2017]) to handle several objects, we encode instance identity as an additional Hough dimension rather than only filtering voters, and we explicitly design for segmentation failures with a soft gating rule, which we evaluate against a strict variant.

## 3 Problem Statement

Given two images $I_1, I_2$ and a set of putative correspondences $\mathcal{M} = \{(\mathbf{p}_i, \mathbf{q}_i)\}_{i=1}^{M}$, where $\mathbf{p}_i \in I_1$ and $\mathbf{q}_i \in I_2$ are keypoint locations produced by a detector and descriptor matching, a fraction $\varepsilon$ of which are incorrect, we address two tasks.

**Task 1: single-model verification.** Estimate the homography $\mathbf{H}$ relating a planar scene in $I_1$ and $I_2$ and the set of inlier correspondences, when keypoints carry no scale or orientation. Accuracy is measured by the mean corner error $e = \frac{1}{4}\sum_{c=1}^{4} \|\pi(\mathbf{H}\mathbf{c}_c) - \pi(\mathbf{H}^*\mathbf{c}_c)\|$ between the image corners $\mathbf{c}_c$ mapped by the estimate and by the ground truth $\mathbf{H}^*$, where $\pi$ denotes dehomogenisation. We report the percentage of image pairs with $e < 1, 3, 5$ pixels, and study how it changes as $\varepsilon$ grows.

**Task 2: multi-object verification.** The scene contains a background and $K$ objects, each moving with its own homography $\mathbf{H}_0^*, \ldots, \mathbf{H}_K^*$. Recover one hypothesis per motion without producing hypotheses that mix matches from different objects. We measure object recall, the number of false hypotheses per image, and cluster purity (Section 5.2).

## 4 Proposed Method

Figure 1 summarises the pipeline: feature extraction and matching, Hough voting, cluster verification, and optional semantic gating.

### 4.1 Features and matching

We use DISK Tyszkiewicz et al. [2020] (pretrained "depth" weights via Kornia Riba et al. [2020]) as the learned detector and SIFT with RootSIFT normalisation Arandjelović and Zisserman [2012] as the classical baseline, with up to 4000 keypoints per image. Matches are mutual nearest neighbours that also pass Lowe's ratio test (0.8 for SIFT, 0.95 for DISK, whose descriptor distances are less peaked).

### 4.2 Hough voting for a similarity pose

A similarity transform has four parameters: scale $s$, rotation $\theta$ and a 2-D translation. Following Lowe Lowe [2004], translation is not voted as raw $(t_x, t_y)$, which depends on the choice of origin, but as the predicted position of the centre $\mathbf{c}_1$ of $I_1$ in $I_2$:

$$\mathbf{c}_2 = \mathbf{q}_a + s\,\mathbf{R}(\theta)\,(\mathbf{c}_1 - \mathbf{p}_a), \tag{1}$$

where $(\mathbf{p}_a, \mathbf{q}_a)$ is an anchor correspondence. Every correct match on a rigid planar object predicts the same $\mathbf{c}_2$. Bins are 30° in $\theta$, a factor of 2 in $s$, and $0.25 \cdot \max(h_1, w_1) \cdot s$ in location. Each vote is cast into the two nearest bins along every dimension ($2^4 = 16$ bins) so that a pose near a bin boundary is not split. A cluster's support is the number of *distinct* matches that voted for it.

**Single-match votes (SIFT).** With keypoint scales $\sigma$ and orientations $\phi$, a single match gives $s = \sigma'/\sigma$ and $\theta = \phi' - \phi$, and serves as its own anchor.

**Pairwise votes (learned features).** DISK keypoints have no $\sigma$ or $\phi$. Two matches $i, j$, however, define the segments $\mathbf{u} = \mathbf{p}_j - \mathbf{p}_i$ and $\mathbf{v} = \mathbf{q}_j - \mathbf{q}_i$, and

$$s = \frac{\|\mathbf{v}\|}{\|\mathbf{u}\|}, \qquad \theta = \operatorname{atan2}(\mathbf{v}) - \operatorname{atan2}(\mathbf{u}), \tag{2}$$

which, together with Eq. (1) using match $i$ as anchor, is a complete similarity vote supporting both $i$ and $j$. Each match is paired with its $k$ nearest matched neighbours in $I_1$, since nearby points are likely to lie on the same surface; pairs closer than 10 px are skipped because short segments give unstable angles.

**Making pairwise voting robust.** Naive pairwise voting fails under heavy contamination (Section 5.1). The cause is specific to pairs. A wrong match $j$ that lies near $i$ in $I_1$ lands at an arbitrary position in $I_2$, so $\|\mathbf{v}\| \gg \|\mathbf{u}\|$ and Eq. (2) implies an absurd scale. Because location bins grow with scale, such votes fall into a few very large bins and form fake peaks: at 80% outliers, 88% of all pair votes implied scales outside $[1/8, 8]$ and the winning cluster had a median scale of 15–34 while the true scale was close to 1. We add two rules:

- **Scale prior:** votes with $s \notin [1/8, 8]$ are discarded.
- **Mutual spatial neighbours:** a pair is kept only if $j$ is also among the $2k$ nearest neighbours of $i$ in $I_2$. A rigid object preserves neighbourhoods; a random wrong match almost never does (cf. GMS Bian et al. [2017]).

### 4.3 Cluster verification

A Hough bin is coarse, so each of the top 10 clusters is refined by iterative least squares with no random sampling: fit an affine transform to the cluster, re-collect *all* matches with reprojection error below 10 px, switch to a full homography (direct linear transform, least squares) once at least 8 inliers exist and tighten the threshold to 3 px, and repeat until the inlier set is stable. The cluster with the most final inliers is returned. The procedure is deterministic: the same input always gives the same output.

### 4.4 Semantic-gated voting

We run a COCO-pretrained Mask R-CNN He et al. [2017] (score $> 0.5$) on both images to obtain instance maps $\iota_1, \iota_2$ (0 = background) with class labels $c(\cdot)$. Each match $m$ receives instance ids $\iota_1(\mathbf{p}_m), \iota_2(\mathbf{q}_m)$. Three rules restrict voting:

- **R1 (class consistency):** discard a match whose endpoints lie on objects of different classes.
- **R2 (same instance):** a pair may vote only if both matches lie on the same instance in $I_1$ and in $I_2$.
- **R3 (instance dimension):** the instance id is appended as a fifth Hough coordinate, so votes from different objects never share a bin, and verification of a cluster only collects inliers from its own instance.

Table 1: HPatches, no injected outliers (580 pairs). Percentage of pairs with mean corner error below 1/3/5 px, split by illumination (Illum) and viewpoint (View) scenes at 3 px, and median verification time.

| Features | Verifier | @1px | @3px | @5px | Illum@3 | View@3 | ms |
|---|---|---|---|---|---|---|---|
| SIFT | RANSAC | 47.9 | 77.2 | 86.6 | 89.5 | 65.4 | 0.6 |
| SIFT | MAGSAC++ | 52.6 | 81.4 | **89.0** | 90.2 | **72.9** | 0.4 |
| SIFT | Hough, single Lowe [2004] | **53.1** | **81.6** | 88.1 | **90.5** | **72.9** | 15.5 |
| SIFT | Hough, pairwise (ours) | 52.6 | 80.7 | 87.2 | 89.8 | 71.9 | 34.3 |
| DISK | RANSAC | 42.8 | 74.7 | 85.0 | 91.9 | 58.0 | 0.9 |
| DISK | MAGSAC++ | 53.8 | 82.6 | **90.0** | 96.5 | **69.2** | 1.1 |
| DISK | Hough, pairwise (ours) | **55.2** | **82.9** | 88.6 | **97.2** | **69.2** | 128.9 |

**Strict vs. soft gating.** In *strict* mode background is an ordinary label, so a match from a detected cup to an undetected region is rejected by R1. In practice detections are often missed in one of the two images, and strict gating then deletes real objects. In *soft* mode, background means "unknown": R1 rejects only when both endpoints are detected *and* disagree, R2 treats an unknown endpoint as compatible, and the group used by R3 is the instance in $I_1$.

### 4.5 Multi-object extraction and duplicate merging

For multi-object scenes, hypotheses are extracted greedily: clusters are visited in order of support (up to 300 clusters), each is refined as above using only matches not yet claimed (and, with gating, only matches of the cluster's instance group), and a hypothesis with at least 8 inliers claims its matches. Gating splits a single motion into several groups when the segmenter detects several "stuff" instances on the background (a table, a wall), producing duplicate hypotheses for the same motion. We therefore *merge* a hypothesis into an earlier one whenever the earlier homography already explains its inliers (median transfer error below 5 px) and refit on the union. Merging is applied identically to every method in our experiments.

## 5 Experiments and Results

**Implementation.** Python with OpenCV, PyTorch and Kornia, on a single NVIDIA GTX 1650 (4 GB) under WSL2 with 4.8 GB RAM. All Hough parameters are fixed across experiments: Lowe's bin sizes, $k = 16$, scale prior $[1/8, 8]$, 3 px final inlier threshold. Baselines are OpenCV's RANSAC and USAC-MAGSAC (MAGSAC++), both with a 3 px threshold, 10,000 iterations and 0.999 confidence.

### 5.1 Single-model verification on HPatches

HPatches Balntas et al. [2017] contains 116 planar scenes (57 with illumination changes, 59 with viewpoint changes) of 6 images each, giving 580 pairs with ground-truth homographies. Images are resized to at most 1024 px. To test robustness we inject synthetic outliers: random correspondences with uniformly random positions (and, for SIFT, random scales and orientations) appended until a fraction $\varepsilon$ of all matches are wrong.

**Results without injected outliers.** Table 1 shows that Hough verification is fully competitive on clean data. With DISK, our pairwise verifier is the most accurate at 1 px and 3 px and on illumination changes, and ties MAGSAC++ on viewpoint changes, while plain RANSAC trails by 8–13 points. With SIFT, pairwise voting (52.6/80.7) is within one point of Lowe's single-match voting (53.1/81.6), which shows that discarding keypoint scale and orientation costs essentially nothing once pairs are used. Our pure-Python implementation is, however, 40–120$\times$ slower than OpenCV's optimised C++ estimators on clean data.

**Robustness to outliers.** Figure 2 shows the main result. Up to 80% outliers all verifiers stay within a few points of their clean accuracy. At 90%, RANSAC drops to 52–60% while MAGSAC++ and pairwise Hough remain at 74–81%. At 95%, MAGSAC++ falls to 36.9% (SIFT) and 33.8% (DISK),

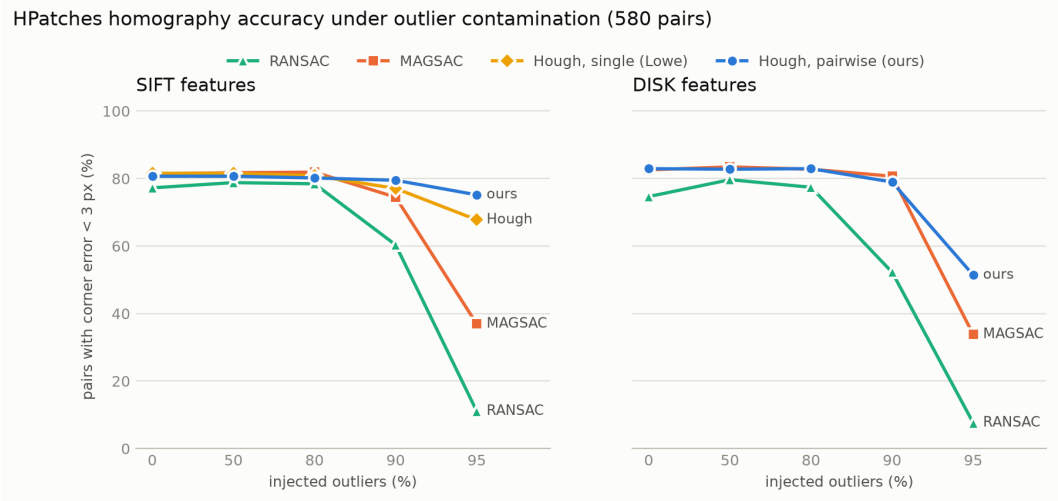



Figure 2: HPatches accuracy (pairs with corner error $< 3$ px) as synthetic outliers are injected. RANSAC collapses above 80%; MAGSAC++ above 90%. Pairwise Hough degrades gracefully.

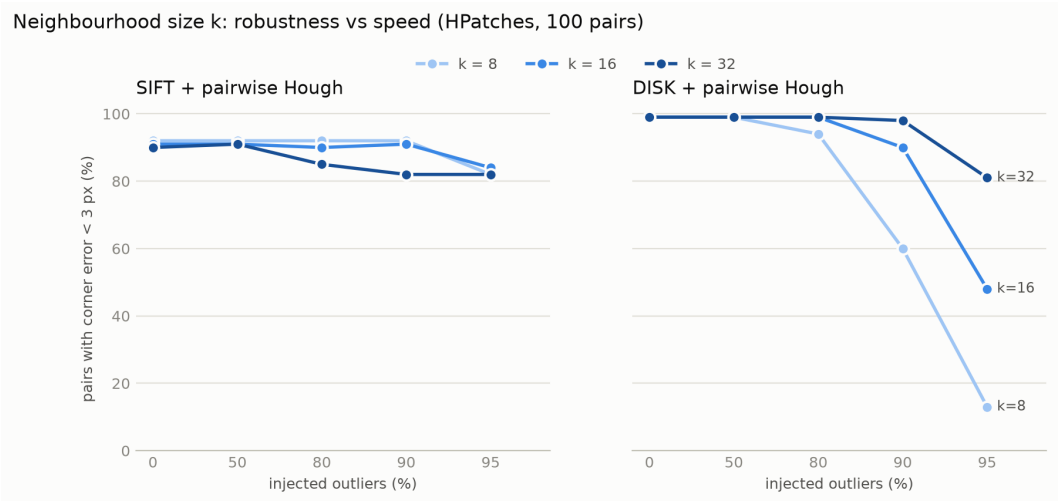



Figure 3: Sensitivity of pairwise Hough to the neighbourhood size $k$ (first 20 HPatches scenes, 100 pairs).

whereas pairwise Hough retains 75.2% and 51.4%. With SIFT, pairwise voting even overtakes Lowe's single-match voting at 95% (75.2% vs. 67.8%): the mutual-neighbour rule removes votes from random matches that single-match voting has no way to reject.

**Why naive pairwise voting failed.** Before adding the scale prior and the mutual-neighbour rule, DISK with pairwise Hough fell from 83.3% at 50% outliers to 13.8% at 80%. Inspecting the accumulator showed that 88% of pair votes implied scales outside $[1/8, 8]$ and that the winning cluster had a median implied scale of 15–34, against a true scale near 1. The two rules raised accuracy at 80% outliers back to 82.9%.

**Sensitivity to the neighbourhood size $k$.** Each match is paired with its $k$ nearest neighbours, so the expected number of *correct* partners per correct match is roughly $k(1-\varepsilon)$. Figure 3 confirms this: SIFT, with few matches per image, is insensitive to $k$, while DISK at 95% outliers (about 35,000 injected matches per pair) needs a wide neighbourhood, improving from 13% at $k = 8$ to 48% at $k = 16$ and 81% at $k = 32$. Larger $k$ costs time (DISK at 95%: 0.15 s, 0.36 s and 2.0 s per pair). We use $k = 16$ throughout as a compromise; it matches the best setting up to 80% outliers.

Table 2: Multi-object test split (150 COCO scenes). Object recall (%, higher is better) and wrong hypotheses per image (lower is better) at 0, 50 and 80% injected outliers; median time at 80%. Sem = semantic gating, +M = duplicate merging.

| | | 0% | | 50% | | 80% | | |
|---|---|---|---|---|---|---|---|---|
| Features | Method | Recall | Wrong | Recall | Wrong | Recall | Wrong | ms |
| DISK | RANSAC | **85.4** | 0.43 | 57.2 | 2.03 | 19.0 | 6.37 | 1353 |
| DISK | Hough | 78.2 | 1.85 | 69.4 | 2.83 | 58.4 | 4.37 | 112 |
| DISK | Sem+RANSAC | 79.6 | 3.13 | **75.8** | 3.30 | 66.9 | 6.79 | 1730 |
| DISK | Sem+Hough | 74.6 | 2.26 | 73.8 | 2.44 | **74.2** | 2.70 | 96 |
| DISK | Sem+RANSAC+M | 79.0 | **0.41** | 74.2 | **0.75** | 66.1 | 4.27 | 1732 |
| DISK | Sem+Hough+M (ours) | 74.3 | 0.58 | 73.2 | 0.83 | 72.7 | **1.05** | 98 |
| SIFT | RANSAC | **78.0** | **0.15** | 60.7 | 0.81 | 22.9 | 3.61 | 791 |
| SIFT | Hough | 73.3 | 1.23 | 69.3 | 1.35 | 47.6 | 2.66 | 79 |
| SIFT | Sem+RANSAC | 74.4 | 1.50 | **72.9** | 1.64 | **67.7** | 2.21 | 874 |
| SIFT | Sem+Hough | 71.9 | 1.33 | 71.1 | 1.37 | 63.1 | 1.42 | 57 |
| SIFT | Sem+RANSAC+M | 73.4 | 0.19 | 72.3 | **0.33** | 67.3 | 0.92 | 875 |
| SIFT | Sem+Hough+M (ours) | 71.1 | 0.34 | 70.6 | 0.43 | 63.1 | **0.86** | 58 |

### 5.2 Multi-object verification

HPatches scenes are single planes and cannot show the effect of semantic gating. We therefore build a multi-object benchmark from COCO val2017 Lin et al. [2014]. For each image with at least two non-overlapping objects covering 3–25% of the image, we take up to three objects with their ground-truth masks. The background is moved by a random camera homography $\mathbf{H}_0$ (rotation $\pm 10°$, scale 0.9–1.1, translation $\pm 5\%$). Each object additionally receives its own motion (rotation 5–20°, scale 0.85–1.15, translation 10–25% of the image size), so $\mathbf{H}_i = \mathbf{T}_i \mathbf{H}_0$. The original object regions are inpainted and the moved objects are pasted on top, so every pixel of $I_1$ has a known motion. Semantic priors come from Mask R-CNN run on the images, never from the COCO masks. The first 150 eligible images (in a fixed random order) form the *test* split and the next 100 a disjoint *validation* split; every design decision below was taken on the validation split, and the test split was evaluated once with frozen settings. Outliers are injected as on HPatches.

A hypothesis is *correct* if its homography agrees with the true motion of the majority label of its inliers (mean error below 5 px over that region) and that motion was not already claimed. Otherwise it is a *duplicate* (correct motion, already found) or *wrong* (no true motion matches, typically a cluster mixing objects or injected outliers). We report object recall (background excluded) and wrong hypotheses per image, for a $2 \times 2 \times 2$ design: verifier (sequential RANSAC or Hough), semantic gating (off or on; for RANSAC, run separately inside each semantic group), and duplicate merging (off or on).

**Results.** Table 2 and Figure 4 show three regimes. *Without injected outliers*, plain sequential RANSAC is the best method for both features: each object has many matches, which is the setting RANSAC handles best, and every semantic variant loses some recall because Mask R-CNN occasionally assigns different classes to the same object in the two images. *At 50% outliers*, the semantic methods lead and Sem+RANSAC+M and Sem+Hough+M are close (DISK: 74.2 vs. 73.2 recall, 0.75 vs. 0.83 wrong), with Hough about $11\times$ faster. *At 80% outliers*, the combination of gating and voting is clearly best with DISK: Sem+Hough+M recovers 72.7% of objects with 1.05 wrong hypotheses per image, against 19.0%/6.37 for RANSAC, 58.4%/4.37 for Hough alone and 66.1%/4.27 for Sem+RANSAC+M, at $17\times$ lower cost than the latter. With SIFT, Sem+RANSAC keeps a higher recall at 80% (67.7 vs. 63.1) while the numbers of wrong hypotheses are similar (0.92 vs. 0.86).

The ablation explains the result. Gating alone (Hough $\rightarrow$ Sem+Hough) makes DISK recall almost independent of the outlier rate (74.6, 73.8, 74.2%) because injected matches rarely satisfy the same-instance pair rule, but it leaves duplicates (0.78 per image at 0%). Merging alone barely changes ungated methods (Hough $\rightarrow$ Hough+M: 1.85 $\rightarrow$ 1.57 wrong at 0%), but after gating it reduces wrong hypotheses by 1.7–3.9$\times$, because many "wrong" hypotheses of gated Hough are partial fits of a

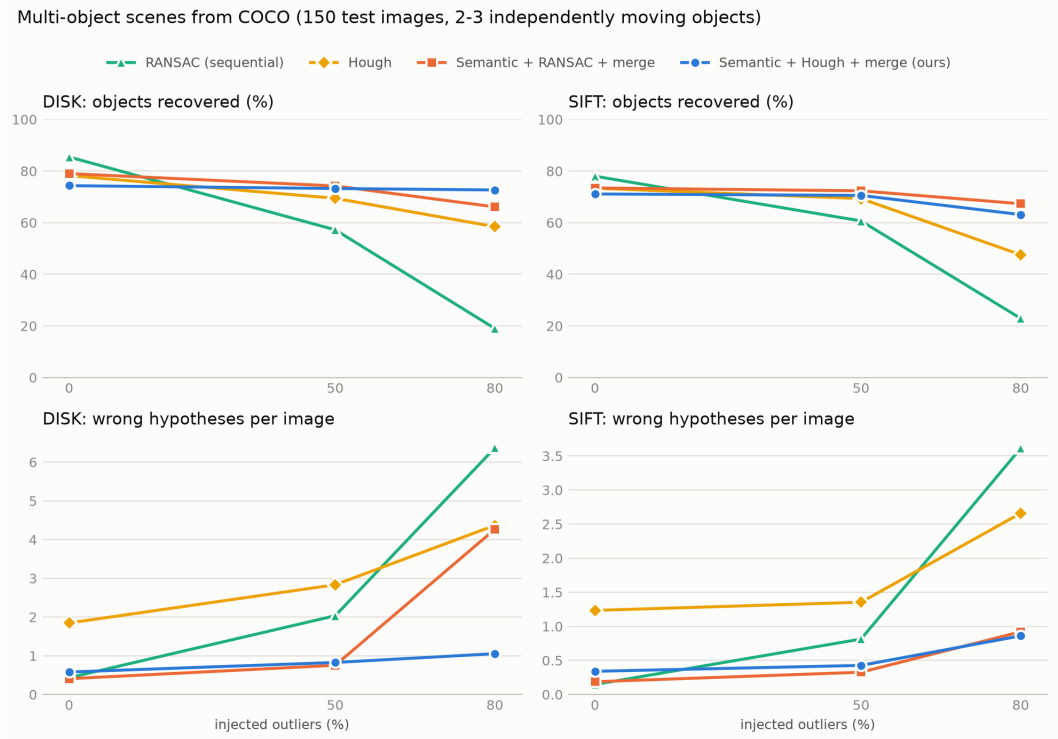



Figure 4: Multi-object test split: object recall (top) and wrong hypotheses per image (bottom) as outliers are injected.

motion that another group already explains. Gating also raises cluster purity, the share of a hypothesis' inliers that belong to one object, from 92.8% to 98.1% on clean DISK scenes.

### 5.3 Variants that did not work

Several design decisions came from failures, which we report for completeness.

- **Strict gating.** Treating background as an ordinary class rejects every match from a detected object to a region where the detector missed it. On the test split (no injected outliers) strict gating reached 69.6% DISK recall against 74.6% for soft gating.
- **Top-60 cluster cap.** An early version visited only the 60 largest clusters. Gating splits background "stuff" into many large clusters, which pushed small objects beyond that cap: 69% of the objects lost by gating had a valid cluster ranked below 60. Raising the cap to 300 for all Hough variants raised gated recall from 60.6% to 74.6%.
- **Mask-transfer verification.** We tested rejecting a hypothesis when warping its object's mask from $I_1$ with $\mathbf{H}$ does not overlap a same-class instance in $I_2$ (IoU $\geq \tau$). On the validation split it cost 4–5 points of recall for every method even at $\tau = 0.05$, while barely reducing wrong hypotheses, because moved objects are often given a different class in $I_2$ (R1 alone rejects 9.8% of correct matches). We therefore did not use it.
- **Absolute numbers.** Mask R-CNN detected 95% of the moved objects, so the remaining recall gap between gated and ungated Hough on clean scenes is caused by class disagreement between the two images and by the rules themselves (14% of correct pair votes are removed by gating), not by missed detections.

## 6 Conclusions

We made Lowe-style Hough pose clustering work with orientation-free learned features by voting with pairs of matches, and made pairwise voting robust by diagnosing and removing the absurd-scale votes




(a) Success. (b) Failure.

Figure 5: Two selected test scenes at 80% outliers (DISK); top rows plain Hough, bottom rows ours. Each colour is one hypothesis. (a) Plain Hough returns 6 false hypotheses with 73% purity; ours recovers all three moving objects and the background with none. (b) Three overlapping people are moved and one is rotated by about 20°; overlapping person instances and DISK's lack of rotation invariance make gating group the wrong matches, and our method recovers none of the objects.

created by wrong matches. The resulting deterministic verifier matches MAGSAC++ on HPatches up to 90% outliers and is clearly more accurate at 95%. For scenes with several independently moving objects, semantic gating and Hough voting are complementary: gating removes the cross-object pairs that random outliers create, and voting tolerates the outliers that remain, so with duplicate merging the combination keeps DISK object recall between 72.7% and 74.3% from 0% to 80% outliers with few wrong hypotheses, where sequential RANSAC falls to 19%. On clean scenes, however, plain RANSAC remains the most accurate multi-object method, and on single planes our verifier is slower than the optimised C++ baselines. The advantage of the approach is therefore specific to heavily contaminated settings.

**Limitations.** DISK, like SuperPoint, is not rotation invariant: in a synthetic test it fell from 47 matches at 0° in-plane rotation to 10 at 35°, while SIFT was unaffected. Our pure-Python Hough implementation is 40–120× slower than OpenCV's C++ estimators on clean data. The outliers are injected synthetically and the multi-object benchmark is synthetic (pasted objects, planar motions). The semantic prior is limited to the 80 COCO classes, and overlapping instances of the same class remain a failure case (Figure 5b).

**Future work.**

1. **Identical-instance recognition.** Scenes with many copies of one object (retail shelves, bin picking) are a known failure case for Hough and RANSAC. Grouping votes by the instance in $I_2$ targets this case directly.
2. **Open-world gating.** Replacing Mask R-CNN with class-agnostic Segment Anything Kirillov et al. [2023] masks removes the 80-class limit, keeping rules R2 and R3.
3. **Learned voting.** A differentiable version of the 16-bin soft accumulator would let a network learn per-match vote weights end to end, with the present pipeline as its non-learned baseline.

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